

INTERN AND PLACEMENT OFFICE SYSTEM (IPOS)

Complete Data, Class and Behavioral Modeling Assignment

Software Requirements Specification and Analysis

Submitted To

Submitted By

Roll / Batch

Submission Date

Modeling rules follow the supplied course notes and the given IPOS problem statement.

1. Problem Statement Summary

The Intern and Placement Office (IPO) of XYZ Institute manages the annual internship placement of BSSE 4th Year students. The internship semester runs from 1 January to 30 June each year.

Interested software companies send their willingness to take interns to the IPO Head. IPOC Member 1 records company information, available job positions, responsibilities, company address and HR email address. IPOC Member 2 informs students to submit their company/job choices in priority order. Students also upload a CV and submit skill information with competency rank, where rank 1 is the highest competency.

IPOC Member 3 records each student's CGPA up to Third Year First Semester. Allocation considers CGPA order, the student's preferences, skill match and position availability. The IPO Head may manually place a student in any company.

After allocation, a formal selection letter is generated for company HR and sent by email on behalf of the IPO Head. A copy is sent to the Regular Program Officer for office record. Companies issue offer letters to selected interns and send a copy to the IPO Head.

2. Modeling Rules Used

- GC: a noun is a potential class if it meets at least 3 of 7 General Characteristics, or if it is GC1 (External Entity).
- SC: if a potential class satisfies SC6 (Essential Requirements), it is selected as a class.
- GC is a gate: nouns that fail GC do not proceed to Selection Criteria.
- Final analysis class set: IPO Personnel, Student, Company, Student Preference, Job Position, Skill, Allocation, Email.

Part A - Data-Based Modeling

A1. Noun Identification

No.	Noun / Noun Phrase	Class.
1	IPOS	SS
2	XYZ Institute	PD
3	IPO	PD
4	IPOC Member 1	SS
5	IPOC Member 2	SS
6	IPOC	PD
7	IPO Head	SS
8	Student	SS
9	Company	SS
10	HR	SS
11	Regular Program Officer	SS
12	Internship	PD
13	Year	PD
14	Semester	PD

No.	Noun / Noun Phrase	Class.
15	Notice	SS
16	Email	SS
17	Formal Selection Letter	SS
18	Offer Letter	SS
19	CV	SS
20	CGPA	SS
21	Skill Set	SS
22	Office Record / Copy	SS
23	Competency Rank	SS
24	Company Choice Form	SS
25	Company Address	SS
26	HR's Email Address	SS
27	Job Position	SS
28	Job Responsibility	SS
29	Priority	SS
30	Authority	PD
31	Company Interest	PD
32	Preference	PD
33	Selection Process	PD
34	Allocation	SS
35	Company Name	SS
36	IPOC Member 3	SS

A2. Group and Reduce

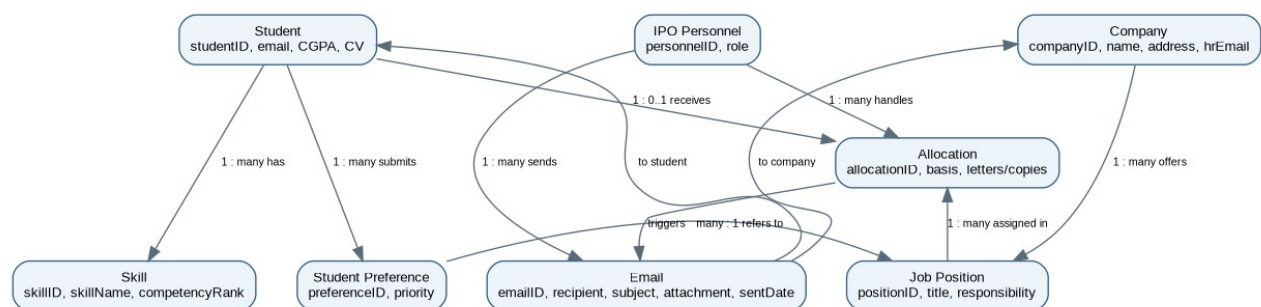
Data Object	Merged / Retained Information
Student	CGPA, CV, student email
Company	company name, company address, HR email
Job Position	job responsibility, availability

Data Object	Merged / Retained Information
IPO Personnel	IPO Head, IPOC members 1-3, RPO; role
Student Preference	company choice form, priority
Skill	skill set, competency rank
Allocation	formal selection letter, office copy, offer letter, allocation basis
Email	sender, recipient, subject, attachment, sent date

A3. Relationships

Relationship	Cardinality	Meaning
Company - Job Position	1 : many	A company offers one or more internship positions.
Student - Student Preference	1 : many	A student submits ordered job-position preferences.
Job Position - Student Preference	1 : many	A job position can appear in many students' preference lists.
Student - Skill	1 : many	A student records multiple skills with competency ranks.
Student - Allocation	1 : 0..1	A student receives at most one final placement for the internship cycle.
Job Position - Allocation	1 : many	A position may receive allocations while availability remains.
IPO Personnel - Allocation	1 : many	IPO personnel handle placements; IPO Head can manually override.
IPO Personnel - Email	1 : many	IPO communication is sent on behalf of authorized personnel.
Email - Student / Company	many : 0..1 each	Emails are directed to students or company HR as required.
Allocation - Email	1 : many	A confirmed allocation can trigger formal placement communication.

A4. ER Diagram



A5. Schema

Table	Attributes
IPO_Personnel	personnelID (PK), role
Student	studentID (PK), emailAddress, CGPA, CVReference
Company	companyID (PK), companyName, companyAddress, hrEmailAddress
Job_Position	positionID (PK), companyID (FK), positionTitle, jobResponsibility, availabilityStatus
Student_Preference	preferenceID (PK), studentID (FK), positionID (FK), priority
Skill	skillID (PK), studentID (FK), skillName, competencyRank
Allocation	allocationID (PK), studentID (FK), positionID (FK), personnelID (FK), allocationBasis, selectionLetterReference, officeCopyReference, offerLetterReference
Email	emailID (PK), senderPersonnelID (FK), studentID (FK, nullable), companyID (FK, nullable), relatedAllocationID (FK, nullable), recipientAddress, subject, attachmentReference, sentDate

Part B - Class-Based Modeling

B1. General Characteristics

No.	General Characteristic
1	External Entities
2	Things
3	Occurrences / Events
4	Roles
5	Organizational Units
6	Places
7	Structures

GC decision rule: At least 3 of 7 GC characteristics, OR GC1 External Entity.

B2. GC Evaluation

Noun / Noun Phrase	GC Met	Decision	Reason
IPOS	5, 7	Fail	System under construction; not modeled as an analysis class.
IPOC Member 1	1, 4, 5	Pass	External Entity (GC1).
IPOC Member 2	1, 4, 5	Pass	External Entity (GC1).
IPO Head	1, 4, 5	Pass	External Entity (GC1).
Student	1, 4, 5	Pass	External Entity (GC1).
Company	1, 5, 7	Pass	External Entity (GC1).
HR	1, 4	Pass	External Entity (GC1).
Regular Program Officer	1, 4, 5	Pass	External Entity (GC1).
Notice	2, 3	Fail	Only two GC characteristics; not GC1.
Email	1	Pass	External Entity exception: GC1 alone is sufficient.
Formal Selection Letter	2, 7	Fail	Only two GC characteristics.
Offer Letter	2, 7	Fail	Only two GC characteristics.
CV	2, 7	Fail	Only two GC characteristics; therefore SC is not applied.
CGPA	2	Fail	Single information item.

Noun / Noun Phrase	GC Met	Decision	Reason
Skill Set / Skill Form	2, 3, 7	Pass	Structured submitted skill information; three GC characteristics.
Office Record / Copy	2, 7	Fail	Only two GC characteristics.
Competency Rank	2	Fail	Single information item.
Company Choice Form	2, 3, 7	Pass	Structured submitted preference information; three GC characteristics.
Company Address	2, 6	Fail	Only two GC characteristics.
HR's Email Address	2	Fail	Single information item.
Job Position	2, 4, 7	Pass	Thing + role/position + structured record.
Job Responsibility	2	Fail	Attribute-level information.
Priority	2	Fail	Attribute-level information.
Allocation	2, 3, 7	Pass	Placement record/event with structure.
Company Name	2	Fail	Attribute-level information.
IPOC Member 3	1, 4, 5	Pass	External Entity (GC1).

B3. Potential Classes

No.	Potential Class
1	IPOC Member 1
2	IPOC Member 2
3	IPOC Member 3
4	IPO Head
5	Regular Program Officer
6	Student
7	Company
8	HR
9	Email
10	Skill Set / Skill Form
11	Company Choice Form
12	Job Position

No.	Potential Class
13	Allocation

B4. Selection Criteria

No.	Selection Criterion
1	Retained Information
2	Needed Services
3	Multiple Attributes
4	Common Attributes
5	Common Operations
6	Essential Requirements

SC decision rule: SC6 (Essential Requirements) is sufficient for class selection; only GC-pass candidates are evaluated.

B5. SC Evaluation

Potential Class	SC Met	Decision	Reason
IPOC Member 1	1, 2, 4, 5, 6	Selected	SC6: essential workflow role.
IPOC Member 2	1, 2, 4, 5, 6	Selected	SC6: essential workflow role.
IPOC Member 3	1, 2, 4, 5, 6	Selected	SC6: essential workflow role.
IPO Head	1, 2, 4, 5, 6	Selected	SC6: essential authority/manual placement role.
Regular Program Officer	1, 2, 4, 5, 6	Selected	SC6: receives/preserves office copy.
Student	1, 2, 3, 6	Selected	Essential participant with retained profile and services.
Company	1, 2, 3, 6	Selected	Essential participant with retained company/HR information.
HR	1, 4	Reduce / merge	Represent through Company.hrEmailAddress; no independent class retained.
Email	1, 2, 3, 6	Selected	SC6: required communication mechanism.
Skill Set / Skill Form	1, 2, 3, 6	Selected -> Skill	Persistent student skill and competency information.
Company Choice Form	1, 2, 3, 6	Selected -> Student Preference	Persistent ordered preference is the class concept.

Potential Class	SC Met	Decision	Reason
Job Position	1, 2, 3, 6	Selected	Essential for preference, matching and placement.
Allocation	1, 2, 3, 6	Selected	Core placement result/process record.

B6. Analysis and Final Classes

No.	Final Class
1	IPO Personnel
2	Student
3	Company
4	Student Preference
5	Job Position
6	Skill
7	Allocation
8	Email

Analysis actions: IPO workflow roles are merged into IPO Personnel; HR is reduced into Company.hrEmailAddress; Company Choice Form is represented as Student Preference; Skill Set / Skill Form is represented as Skill.

B7. Attributes and Methods

Class	Attributes	Methods
IPO Personnel	- personnellID - role	+ recordCompanyDetails() + recordStudentCGPA() + recordJobPosition() + sendInternshipNotice() + sendEmail() + handleAllocation() + manualPlaceStudent() + preserveOfficeCopy()
Student	- studentID - emailAddress - CGPA - CVReference	+ viewJobPositions() + submitPreference() + uploadCV() + addSkill() + receiveEmail() + receiveAllocation()
Company	- companyID - companyName - companyAddress - hrEmailAddress	+ submitInterest() + provideDetails() + offerJobPosition() + receiveEmail() + issueOfferLetter()
Student Preference	- preferenceID - studentID - positionID - priority	+ createPreference() + setPriority() + submitPreference() + updatePreference() + viewPreference()

Class	Attributes	Methods
Job Position	- positionID - companyID - positionTitle - jobResponsibility - availabilityStatus	+ addPosition() + updatePosition() + viewPosition() + checkAvailability()
Skill	- skillID - studentID - skillName - competencyRank	+ addSkill() + setCompetencyRank() + updateSkill() + viewSkill()
Allocation	- allocationID - studentID - positionID - personnellID - allocationBasis - selectionLetterReference - officeCopyReference - offerLetterReference	+ matchStudent() + allocateStudent() + manualOverride() + generateSelectionLetter() + recordOfficeCopy() + recordOfferLetter()
Email	- emailID - senderAddress - recipientAddress - subject - attachmentReference - sentDate - relatedAllocationID	+ composeEmail() + sendEmail() + recordEmail()

B8. CRC Cards

IPO Personnel

Responsibilities	Collaborators
• Represents IPO Head, IPOC Members 1-3, and Regular Program Officer by role. • Records company/job/student information according to the assigned role. • Handles allocation; IPO Head can override automatic placement. • Sends required communication and preserves office copies.	Student, Company, Job Position, Allocation, Email

Student

Responsibilities	Collaborators
• Maintains the student placement profile. • Submits ranked company/job choices. • Provides CV, skills and competency ranks. • Receives placement communication and allocation result.	Student Preference, Job Position, Skill, Allocation, Email

Company

Responsibilities	Collaborators
• Provides company, HR contact, and internship-position information. • Offers job positions and job responsibilities. • Receives the formal selection communication. • Issues offer letters for selected interns.	IPO Personnel, Job Position, Allocation, Email

Student Preference

Responsibilities	Collaborators
- Records a student-selected job position and priority. - Provides ordered preferences to the allocation process.	Student, Job Position, Allocation

Job Position

Responsibilities	Collaborators
- Represents an internship position offered by a company. - Stores the responsibility and availability used in matching.	Company, Student, Student Preference, Allocation, IPO Personnel

Skill

Responsibilities	Collaborators
- Records a skill held by a student and its competency rank. - Supplies competency information to matching/manual placement.	Student, Allocation

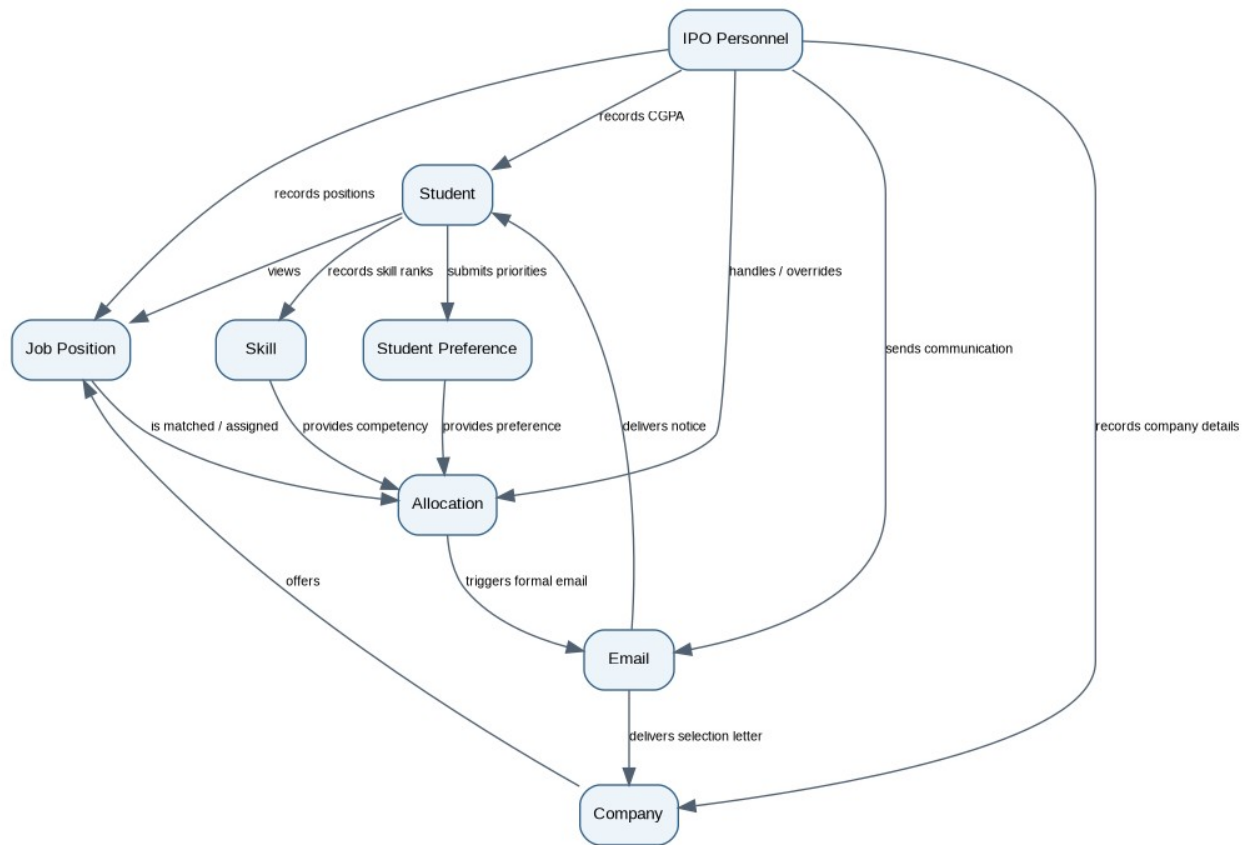
Allocation

Responsibilities	Collaborators
- Matches by CGPA order, preference, skill match and position availability. - Records the final student-to-position placement. - Records whether placement was automatic or manual. - Preserves selection/office/offer document references.	Student, Job Position, Student Preference, Skill, IPO Personnel, Company, Email

Email

Responsibilities	Collaborators
- Carries internship notices and formal placement communication. - Records sender, recipient, subject, attachment and sent date.	IPO Personnel, Student, Company, Allocation

B9. CRC Diagram



Part C - Behavioral Modeling

C1. Event Table

ID	Event	Initiator	Collaborators
E1	Company expresses interest in taking interns.	Company	IPO Personnel
E2	Company and job-position details are recorded.	IPO Personnel	Company, Job Position
E3	Internship notice / communication is sent to students.	IPO Personnel	Email, Student
E4	Student views available company/job information.	Student	Job Position, Company
E5	Student submits company/job preferences with priority.	Student	Student Preference, Job Position
E6	Student uploads CV and submits skills with competency rank.	Student	Skill
E7	Student CGPA is recorded.	IPO Personnel	Student
E8	System matches by CGPA, preference, skill and availability.	Allocation	Student, Student Preference, Skill, Job Position
E9	IPO Head manually places/overrides a student allocation.	IPO Personnel	Allocation, Student, Job Position
E10	Formal selection letter is generated after allocation.	Allocation	Student, Company
E11	Formal communication is emailed to company HR.	Email	IPO Personnel, Company, Allocation
E12	Office copy is sent/preserved for record.	IPO Personnel	Allocation
E13	Company issues an offer letter.	Company	Student, Allocation
E14	Offer-letter copy is preserved by IPO.	IPO Personnel	Company, Allocation

C2. Overall State Transition Model



C3. Per-Class State Models

IPO Personnel

From	Trigger	To
Ready	record details/CGPA	Recording Data
Recording Data	send notice/email	Communicating
Communicating	allocation cycle	Handling Allocation
Handling Allocation	preserve copy / finish	Ready



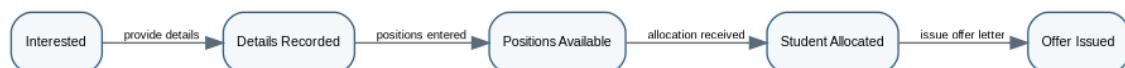
Student

From	Trigger	To
Eligible	submit choice	Preference Submitted
Preference Submitted	CV + skills + CGPA ready	Profile Complete
Profile Complete	allocation result	Allocated
Allocated	offer accepted / internship starts	Internship Active
Internship Active	semester ends	Completed



Company

From	Trigger	To
Interested	provide details	Details Recorded
Details Recorded	positions entered	Positions Available
Positions Available	allocation received	Student Allocated
Student Allocated	issue offer letter	Offer Issued



Student Preference

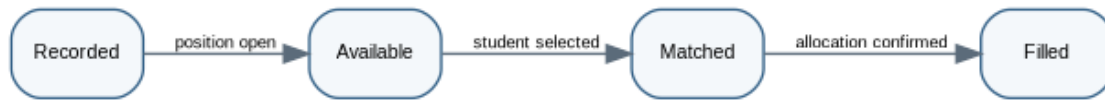
From	Trigger	To
Draft	student submits	Submitted
Submitted	priority validated	Ordered
Ordered	allocation starts	Used in Matching



Job Position

From	Trigger	To
Recorded	position open	Available

From	Trigger	To
Available	student selected	Matched
Matched	allocation confirmed	Filled



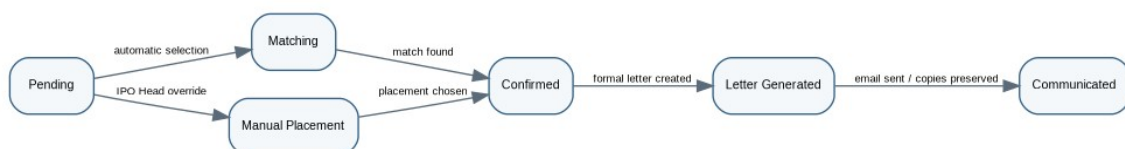
Skill

From	Trigger	To
Recorded	competency rank set	Ranked
Ranked	allocation evaluates skill	Used in Matching



Allocation

From	Trigger	To
Pending	automatic selection	Matching
Pending	IPO Head override	Manual Placement
Matching	match found	Confirmed
Manual Placement	placement chosen	Confirmed
Confirmed	formal letter created	Letter Generated
Letter Generated	email sent / copies preserved	Communicated



Email

From	Trigger	To
Composed	send	Sent
Sent	sent date / attachment stored	Recorded



D. Final Class Set

No.	Final Class
1	IPO Personnel
2	Student
3	Company
4	Student Preference
5	Job Position
6	Skill
7	Allocation
8	Email